

## Monthly Expense Log

Tata, Stuart, and Rere are maintaining a shared monthly expense log. The log is stored as an array where each position represents the amount spent on a given day. Because they keep changing their spending habits, they need to update past entries and also compute the total spending over certain ranges. They decide to ask you to implement the system because they can't trust each other's arithmetic. Your task is to maintain their expense array and answer range-sum queries efficiently.

### Problem Description

You are given an array of  $N$  integers representing daily expenses. Then you must process  $Q$  operations:

- $U\ i\ x$  - Updates expense on day  $i$  to value  $x$ .
- $S\ l\ r$  - Sum the total expense from day  $l$  to day  $r$ .

Operate on a dynamically allocated array (no STL containers).

### Input Format

Line 1:  $N\ Q$ , where  $N$  is the total day and  $Q$  is the total operations. Line 2: the  $N$  integers, representing the expense for each day. Next  $Q$  lines: one operation per line.

### Output Format

For each  $S\ l\ r$ , print the sum.

### Constraints

- $1 \leq N \leq 365$
- $1 \leq Q, x \leq 10^5$
- $1 \leq i, l, r \leq N$

### Sample Input 1 (standard input)

```
5 5
1 2 3 4 5
U 1 10
U 4 14
U 5 3
S 1 2
S 2 5
```

### Sample Output 1 (standard output)

```
12
22
```

### Explanation 1

Here are the updates of the array values from operations above:

```
U 1 10 -> Array = 10 2 3 4 5
U 4 14 -> Array = 10 2 3 14 5
U 5 3 -> Array = 10 2 3 14 3
S 1 2 -> 10 + 2 = 12
S 2 5 -> 2 + 3 + 14 + 3 = 22
```

### Sample Input 2 (standard input)

```
4 6
1 1 1 1
S 1 4
U 1 10
S 1 2
U 4 100
S 3 4
S 1 4
```

### Sample Output 2 (standard output)

4  
11  
101  
112

